

Tips for Reviewing Statistics and Methods for Non-Statisticians

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Disclosures

- StatReviewer Software
 - Co-founder

Outline

- Common misperceptions about peer-review
- Outline for approaching methodological/statistical review
 - Tips and tricks for finding common problems
- Additional help and getting recognition for your good work

Common Automatic Thoughts

Thoughts

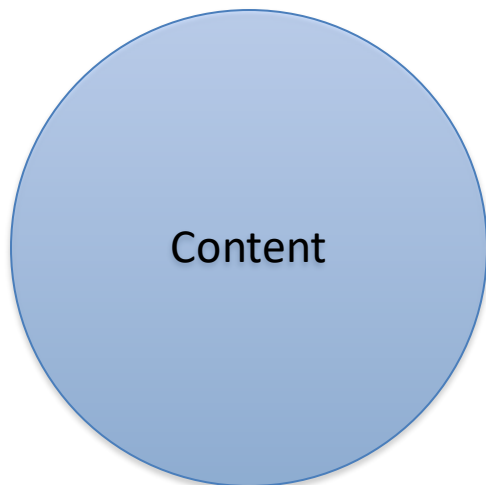
- All-Or-Nothing Thinking
 - If not a Nobel Prize winner than nothing to offer
- Minimization
 - I do not understand ALL of the methods
 - Others would be better at reviewing
- Catastrophizing
 - I will make a fool of myself
 - My review will not be helpful

Counter

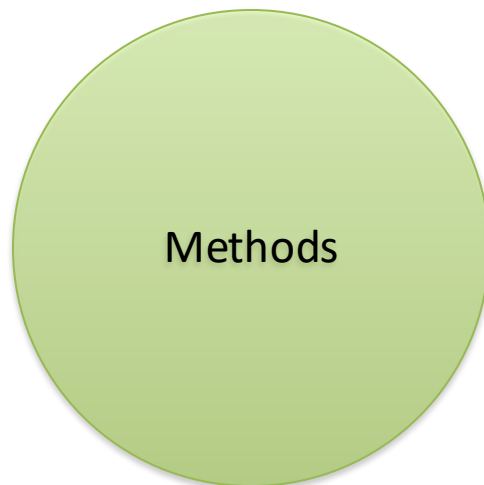
Anyone with content knowledge and/or methods knowledge is valuable

Science is increasingly specialized; common to have teams of experts contribute; not expected to be an expert at all aspects of a study

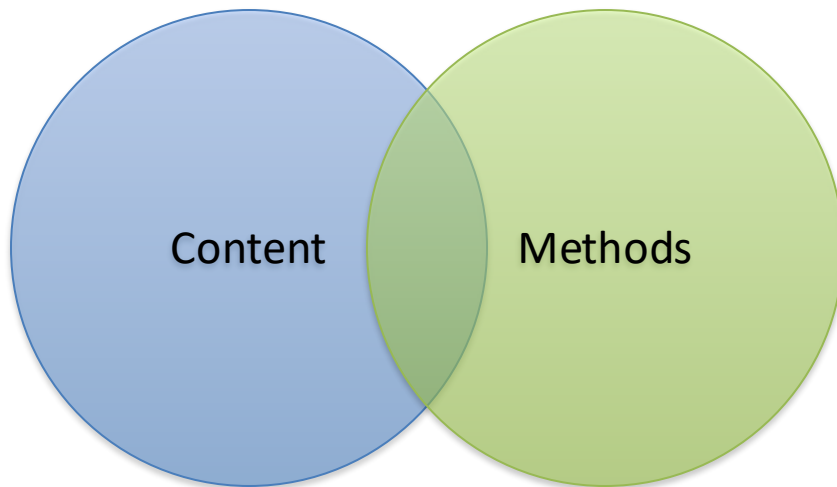
Editors are very grateful for your time; Thoughtful effort is always helpful; Great way to learn



Disease(s)
Procedures
Participants
Clinical issues
Outcomes
Adverse events



Hypotheses
Design
Statistics
Results
Interpretation
Visualization



Content

Methods

Strategy for Methods Review

1. Identify the study design
2. Type of measurements
3. Statistical analysis plan
 - Underlying assumptions
4. Results reporting
 - Effect sizes
 - Common errors
5. Interpretation

Name the Design

- Is it clear what type of study was conducted?
 - Title
 - Abstract
 - Methods
- Can the stated aims/hypotheses be addressed given the design?
- What are the inherent limitations in the design?
- What type of analyses will be needed given the design?

Name the Design

- All empirical research can be placed into one study design type
 - Clinical Trial
 - Observational (Cohort, Case-Control, Longitudinal)
 - Experimental Study (Animal)
 - Systematic Review/Meta-analysis
 - Diagnostic Accuracy Study
- Different systems for categorizing research.

3



Experimental Design

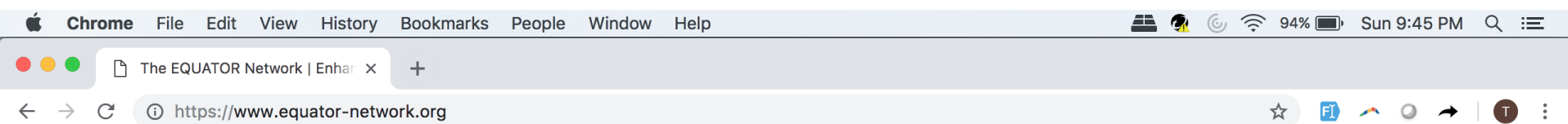
Procedures for the
Behavioral Sciences

Roger E. Kirk

Experimental and Quasi-Experimental Designs for Research

Donald T. Campbell
Julian C. Stanley

Methods Review



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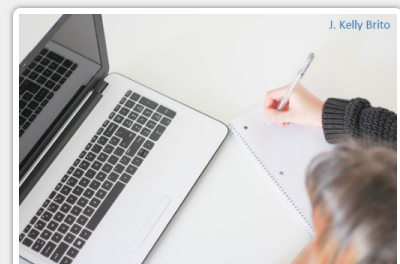
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Type of Measurements

- How are the ideas quantified?
 - Have these measurements been validated previously?
- Do the measurements correspond to the ideas being studied (i.e., validity)?
- Would these measurements be expected to be reproducible (i.e., reliable)?

Level of Measurement

Nominal

Binary: dead/alive, yes/no

Categorical: ethnicity, medical diagnosis

Ordinal: ASA, morbidity scale, attitudes, rating scales

Interval: IQ, Pain ratings

Ratio: age, walking speed, weight

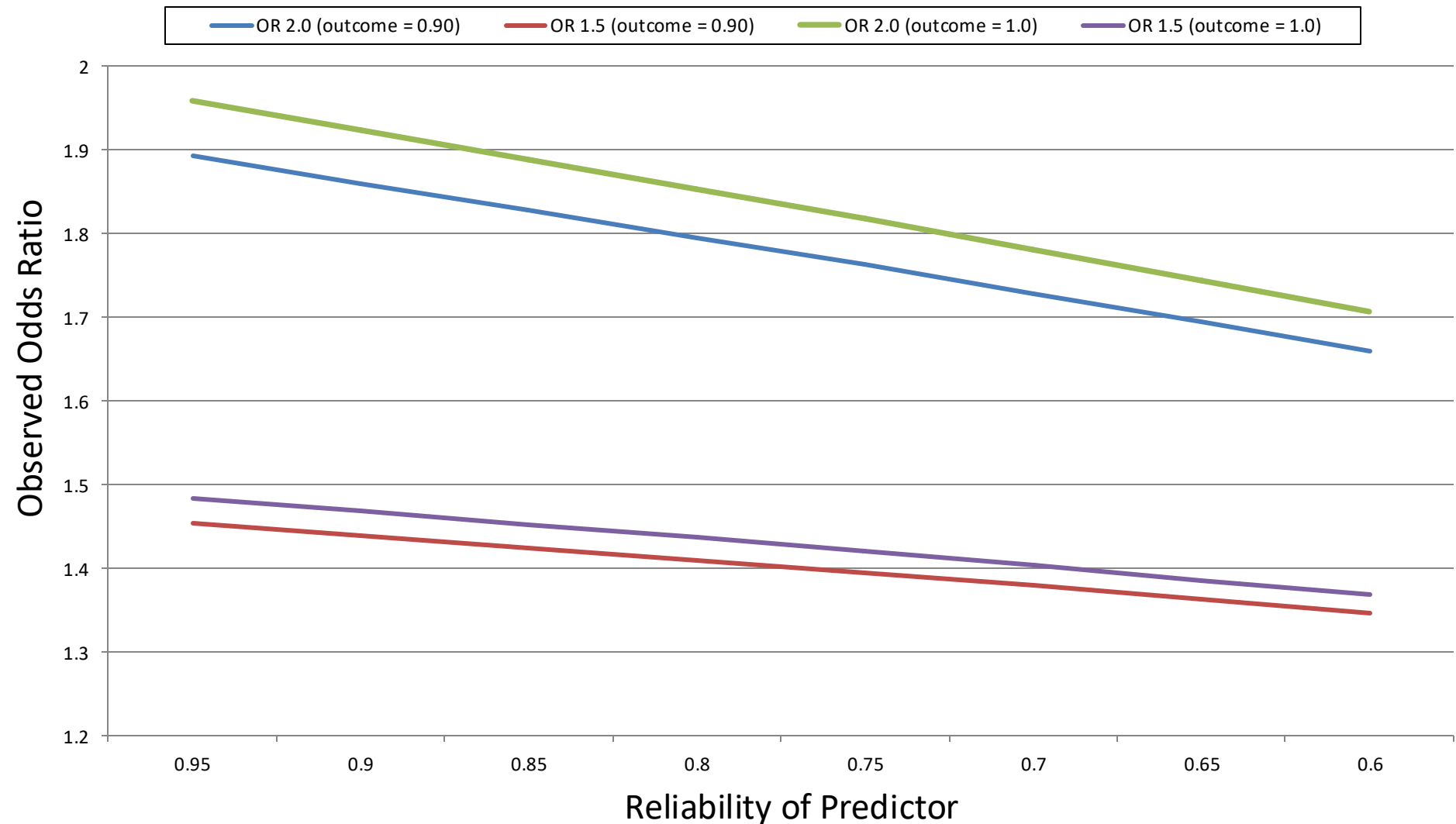
The Reliability of a Variable

- Studies seldom explore the statistical reliability of the measurements
 - Perhaps because the measurements are drawn from a biological sample, it is assumed to have high reliability (i.e., precision).
- However, any measurement can be obscured by error
 - Measurement error (random error)
 - Imprecision of assay, sampling
 - Systematic error (bias)
 - Diurnal variation of a biologic
- Reliability and validity are contextual (i.e., for a specific setting)

$$\text{"True" association}_{xy} = \frac{\text{Observed Association}_{xy}}{\text{Reliability}_x * \text{Reliability}_y} = r_{x'y'} = \frac{r_{xy}}{\sqrt{r_{xx} * r_{yy}}}$$

The Reliability of a Predictor/Outcome

“True” odds ratio (reliability of outcome)



Statistical Analysis Plan

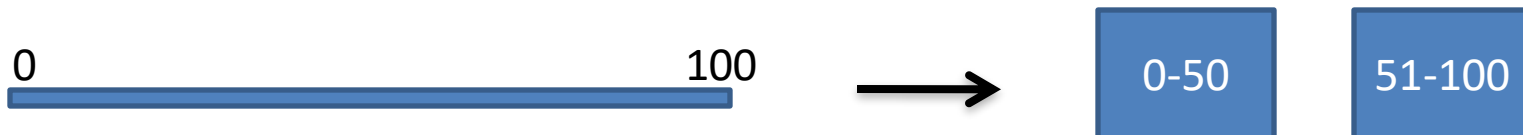
- Does there appear to be a focused plan of analysis?
- How much of the analyses are post hoc (i.e., created after examination of the data)?
- If you had the data, could you run the analyses?
 - Could you ask a statistician to run the analyses given the descriptions?

Statistical Analysis Plan

- Every statistical test requires that certain assumptions have been met
 - Parametric (distribution, homogeneity of variance)
 - Skew, time, missing data, multiple scores per person/unit
- Do the tests correspond to the study design and are they actually used?
 - Self-plagiarism

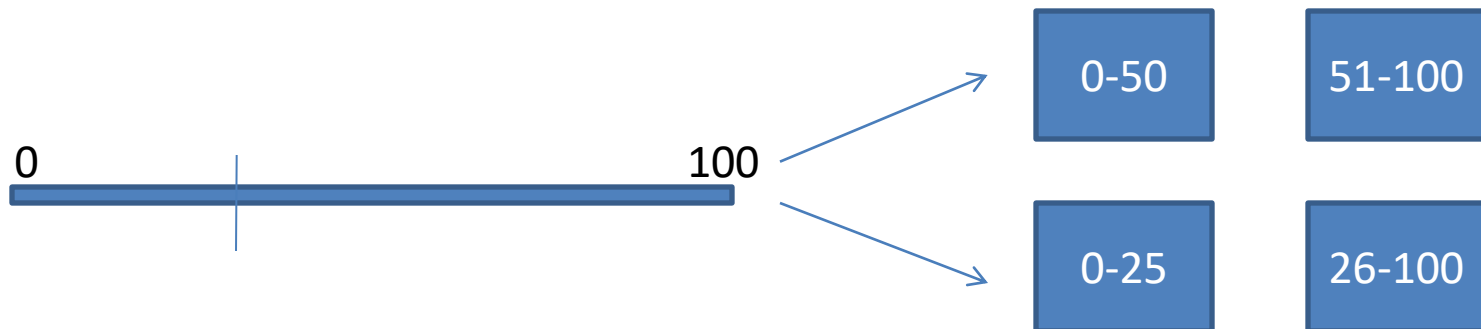
Dichotomization of a variable that is naturally continuous

- Although proven to be a poor practice (1993), medical researchers continue to transform continuous variables (e.g., BMI) into discrete categories for analysis (e.g., underweight, normal, obese)
- For clinical purposes this can make sense, as in a decision point
- However for the statistical evaluation of a biomarker, this choice is quite often a bad idea.



Dichotomization of a variable that is naturally continuous

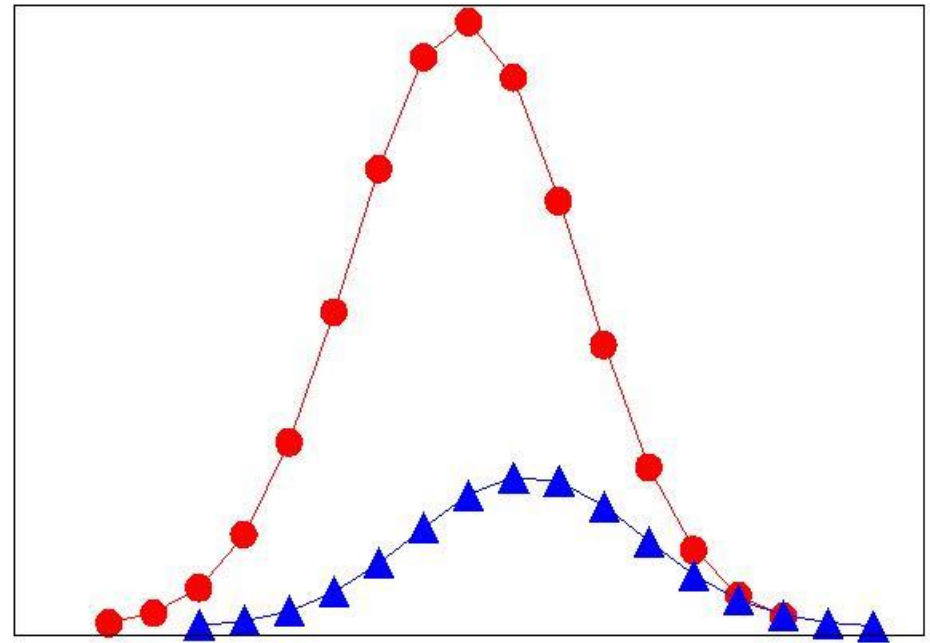
- Why do this?
 - Simplify analysis (yes)
 - Simplify interpretation (yes and no)
 - Dichotomizing is more reliable than using imprecise measurements (no)
 - An “optimal” cut-off can be used (yes, at a cost)



Dichotomization of a variable that is naturally continuous

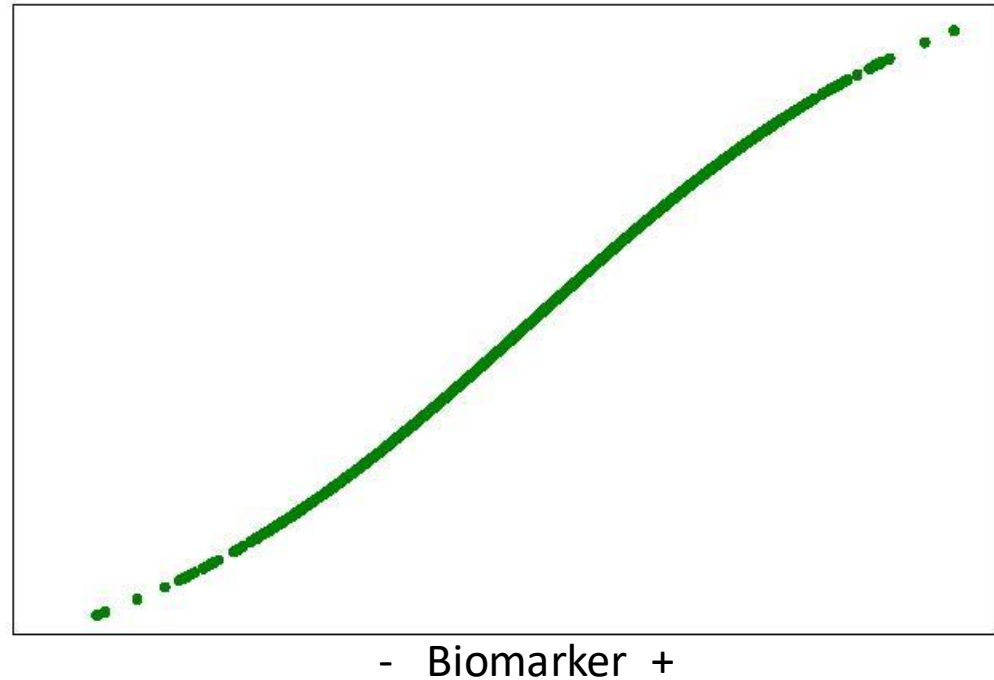
- Why NOT do this?
 - Loss of statistical power
 - When $\sim N(\mu, \sigma)$, analyses are only 65% as efficient
 - Might as well throw away 1/3rd of the data!
 - Loss of information
 - Non-linearity cannot be examined
 - Within-group variability absorbed in the entire group
 - Scores close to the cut-point are viewed inappropriately
 - Dichotomized confounders are not well modeled (residual confounding)
 - Ambiguous choice of the “optimal” cut-off

Dichotomization of a biomarker that is naturally continuous



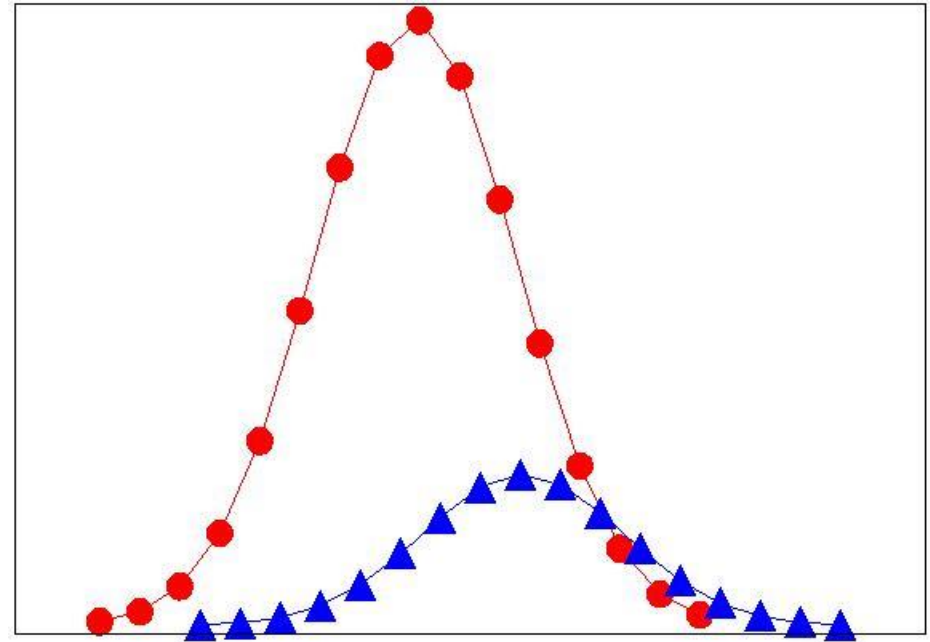
Difference = 0.50 SD

Probability of being diseased

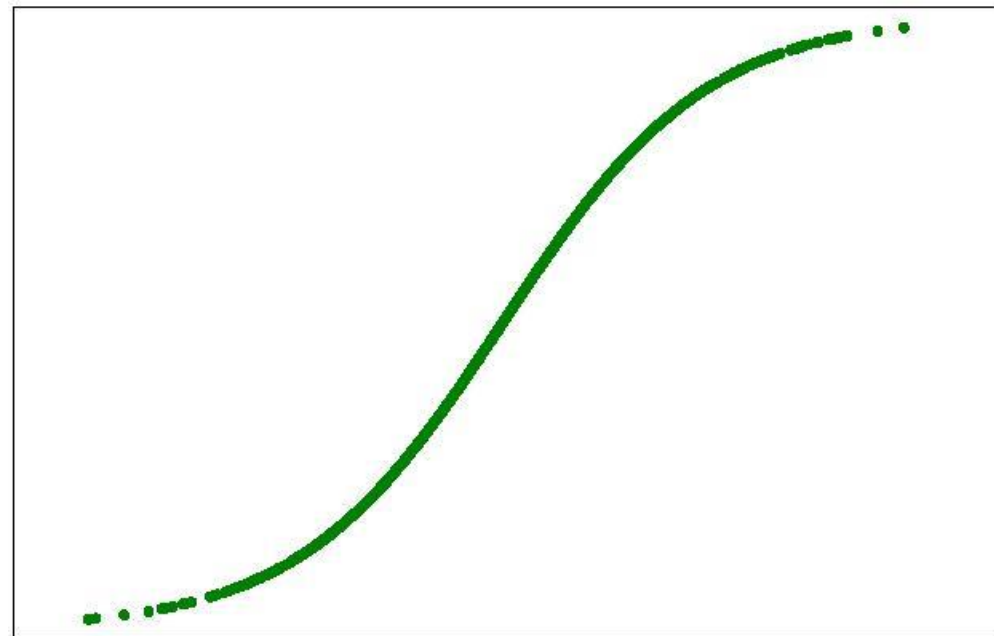


Dichotomization of a biomarker that is naturally continuous

Difference = 1.0 SD

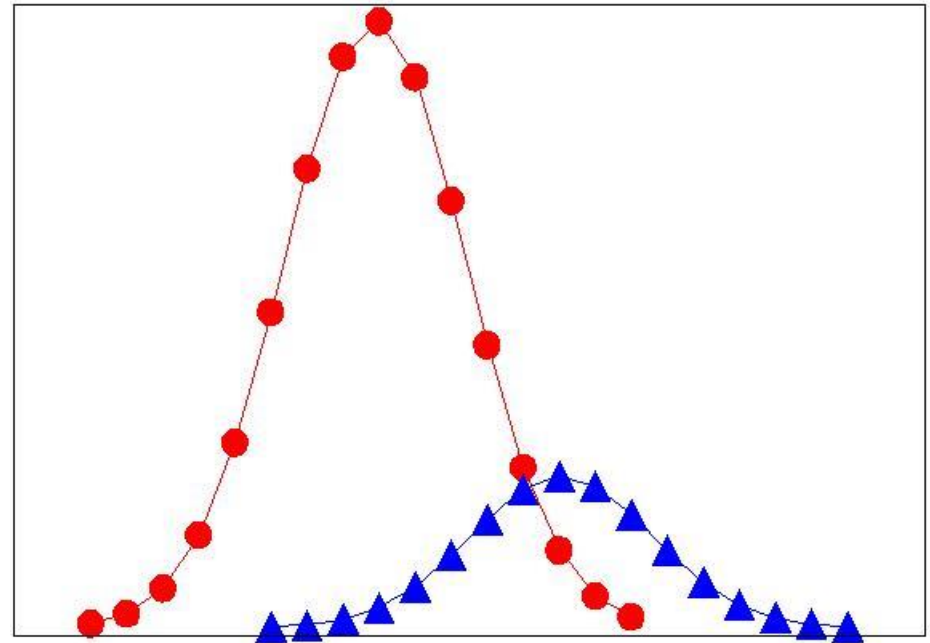


Probability of being diseased

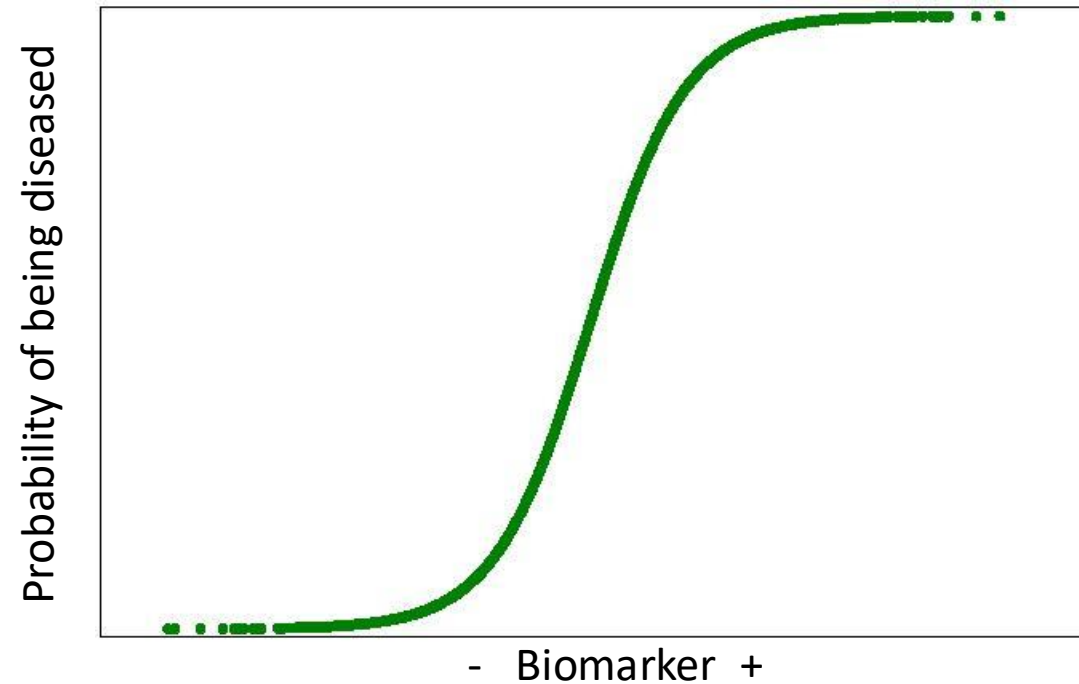


- Biomarker +

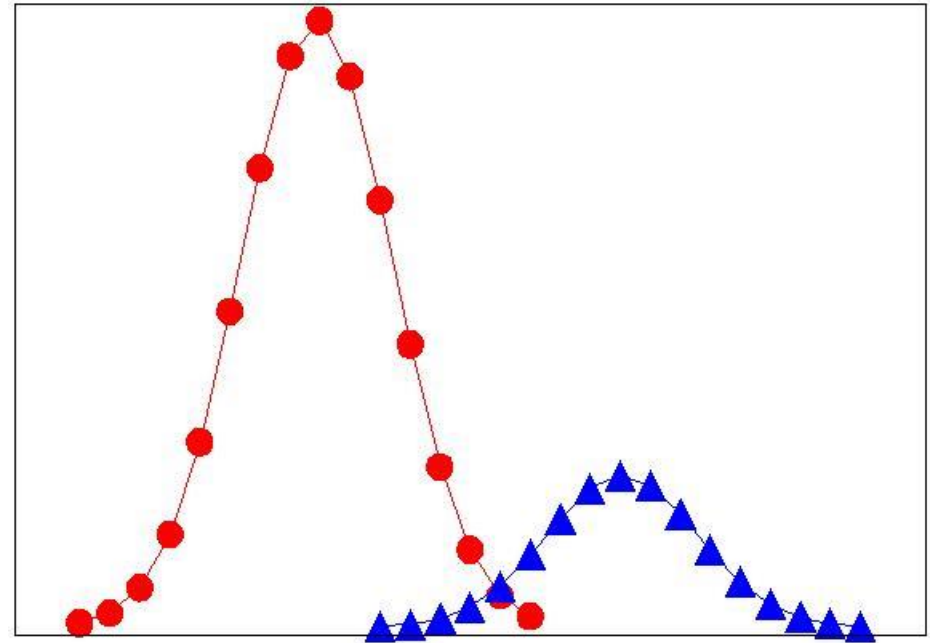
Dichotomization of a biomarker that is naturally continuous



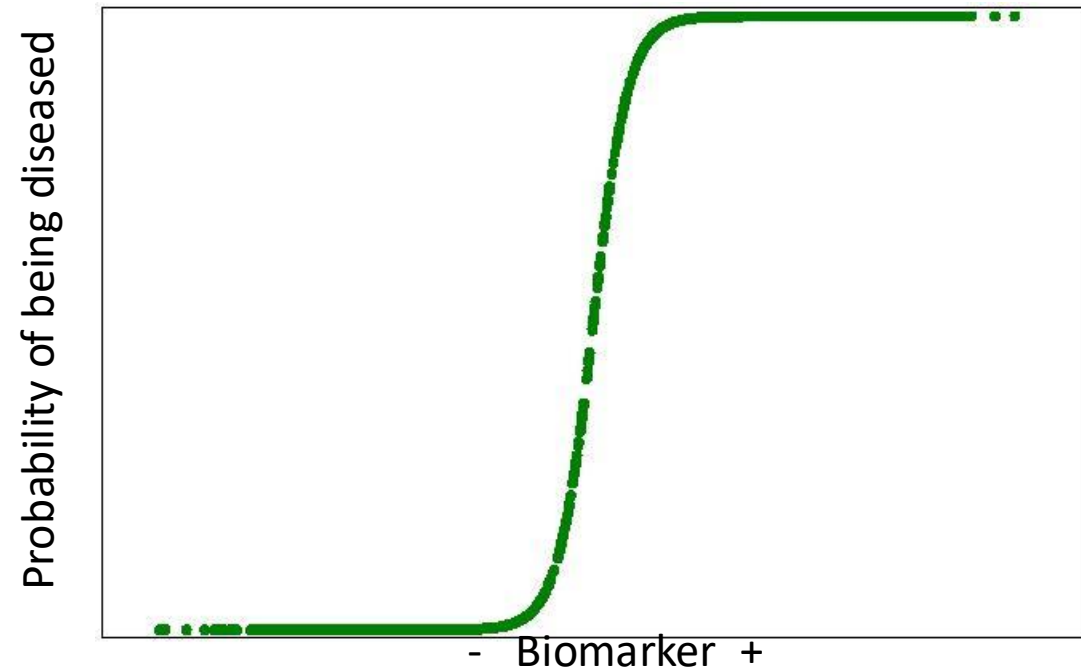
Difference = 2.0 SD



Dichotomization of a biomarker that is naturally continuous



Difference = 4.0 SD



Reporting of Results

- Can you identify the size of the effects?
 - Are more than just p-values reported?
 - Descriptive statistics!
- What is the uncertainty around the findings?
 - Not just a dichotomous finding (statistically significant vs. not)
 - Variability estimates (SD, IQR[25th, 75th])
 - Confidence intervals!

OR 3.5 (95%CI: 3.0 to 4.0)

Common Error: Worshipping the p-value

- Solitary p-value
 - “The control group was inferior to the treatment group on mortality ($p < 0.05$), length of stay ($p < 0.05$), and adverse events ($p < 0.05$)”
- Contrasting p-values
 - “The treatment group was significantly different than baseline ($p < 0.05$), but the control group was not ($p > 0.05$)”
- Failed Superiority $\neq$ Equivalence
 - “The groups were equivalent on the primary outcome, $p = 0.745$ ”

Interpretation

- Does the interpretation reflect the actual data?
 - Primary outcomes treated as primary?
 - Focus on significant findings while ignoring non-significant ones
- Can the study design actually support the interpretations?
 - Convenience sampling cannot yield population prevalence estimates
 - Observational studies cannot be interpreted as randomized designs
 - Beware the pre-test post-test design!

Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

Boutron I, Dutton S, Ravaud P, Altman D. (May, 2010)

- Frequency and degree of “Spin” in published RCTs
- Spin[†]: specific reporting that could distort the interpretation of results and mislead readers.
 - From ignorance of scientific (statistical) issue
 - Unconscious bias
 - Willful intent to deceive

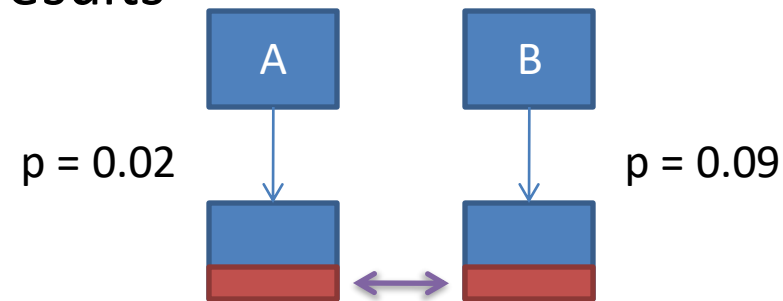
[†]Fletcher RH, Black B. “Spin” in scientific writing: scientific mischief and legal jeopardy. *Med Law.* 2007; 26(3):511-525.

Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

Boutron I, Dutton S, Ravaud P, Altman D. (May, 2010)

Strategies of Spin

- A focus on statistically significant results
 - within-group comparison
 - secondary outcomes
 - subgroup analyses
 - modified population of analyses

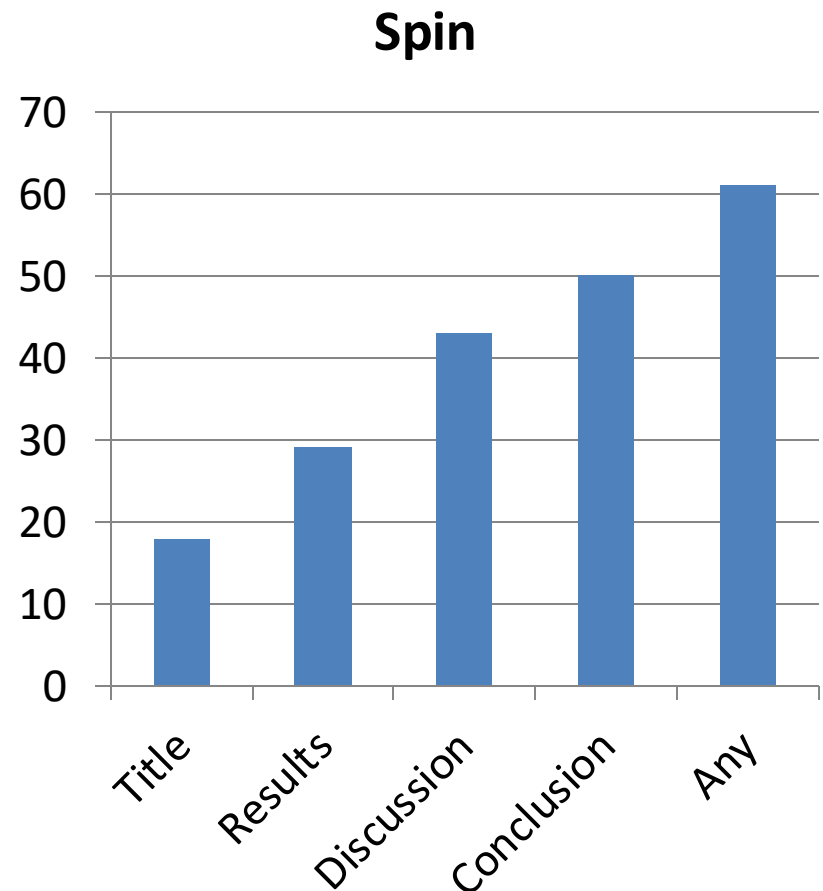


- Interpreting statistically nonsignificant results for the primary outcomes as showing treatment equivalence or comparable effectiveness
- Claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results.

Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

Boutron I, Dutton S, Ravaud P, Altman D. (May, 2010)

- 1735 RCTs identified from December 2006
 - N = 72 Included in analysis
 - All were two group comparisons with nonsignificant primary findings
- 61.1% of manuscripts had spin in at least one section



Next Steps

- Getting more training
- Getting credit for your work!



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Assistant Professor - Anesthesiology and Critical Care, University of Pennsylvania - 2012 to Present

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BIO

I am a board-certified pediatric anesthesiologist and clinical informatician. My clinical expertise includes pediatric and maternal-fetal anesthesia, ultrasound-guided regional anesthesia and vascular access, and pediatric difficult airway management. My research focus is on perioperative informatics and analytics to better understand and improve patient outcomes.

RESEARCH FIELDS

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